CS 203 – Software Tools and Techniques in AI: Project Proposal

Project 5: Cross-Lingual Institute Voice Helpdesk with Drift-Triggered Retraining

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1. Problem and Goal

Helpdesk queries (admissions, hostel/mess, IT, student services) are spoken in code-mixed English–Hindi–Gujarati, and generic ASR degrades on such speech, on new accents, and on new institute vocabulary (course and building names). We will build a lightweight ASR pipeline that transcribes and routes helpdesk queries, **monitors its own quality**, detects silent degradation as the speaker population shifts, and **triggers retraining** before service quality drops.

2. Proposed Approach

- **ASR model:** Fine-tune a small pretrained model (Whisper-small/base, with an AI4Bharat IndicWav2Vec/IndicConformer model as a baseline) on code-mixed helpdesk speech. Compare zero-shot, full fine-tuning, and parameter-efficient (LoRA) fine-tuning on WER/CER, latency, and model size to keep the system lightweight.
- **Routing:** A small text classifier (TF-IDF+LR baseline, then a compact multilingual transformer) maps each transcript to a helpdesk category (admissions, hostel/mess, IT, student services).
- **Drift monitoring:** Label-free signals on live traffic: mean decoder confidence/log-probability, language-ID mix, audio statistics (SNR, duration), and encoder-embedding drift (PSI/MMD/KS test against a reference window). A small, periodically human-verified sample provides true WER as ground truth to calibrate the proxies.
- **Retraining trigger:** If proxy drift or sampled WER crosses a threshold for a sustained window, a pipeline (GitHub Actions/script) fine-tunes on reference plus newly verified data, validates against a fixed test set and the drifted cohort, and promotes the model only if it improves without regressing.
- **Drift simulation:** Speakers are split into cohorts by accent/region and time (e.g., cohort A = training population, cohort B = new "admissions batch"). Cohort B, added noise/codec degradation, and unseen vocabulary are fed as a stream to test detection delay and recovery.
- **MLOps and deployment:** DVC (data versioning), MLflow/Weights & Biases (experiments, model registry), Evidently (drift reports), FastAPI + Docker for serving, and a Gradio/Streamlit demo with a live monitoring dashboard.

3. Data and Sources

- **Self-collected helpdesk speech (primary):** Team members and volunteers from diverse regions record scripted and spontaneous helpdesk-style queries (code-mixed), across phones/laptops and noise conditions. Scripts are seeded from real institute FAQs, mess/hostel notices, IT-support pages, and course/building names from the institute website. Target: several hours from 30+ speakers, with speaker, region, and device metadata (collected with consent, anonymised).
- **Public speech corpora (pretraining, baselines, and augmentation):** AI4Bharat IndicVoices and Kathbath, Mozilla Common Voice (Hindi, Gujarati, Indian English), and the MUCS 2021 Hindi–English code-switching data.
- **Augmentation:** MUSAN/room-impulse-response noise and reverberation, speed/pitch perturbation, and codec compression to mimic low-end devices; optional TTS for new-vocabulary phrases.
- **Labelling (Assignment 1):** Transcription and intent labels using Label Studio with written guidelines (code-mixing, script/normalisation rules); inter-annotator agreement (WER between annotators, Cohen's/Fleiss' κ for intent), disagreement analysis, and a validated held-out test set.

4. Evaluation Plan

WER/CER (overall, per language and per cohort), intent accuracy/macro-F1, real-time factor and model size, drift detection delay and false-alarm rate, and WER recovery after automated retraining.

Note: This proposed solution is not final; the approach may evolve based on implementation, experimentation, and results.